

ECE 332: Engineering Electromagnetics II

Lecture 15

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Chapter 9: Radiation and Antennas

An antenna

- **transmits** – converts a guided EM wave into a wave that propagates in an unbounded medium,
- or **receives** – converts an EM wave propagating in an unbounded medium into a guided EM wave.

The **size** of an antenna is measured in units of the **wavelength** λ it is intended to transmit or receive.

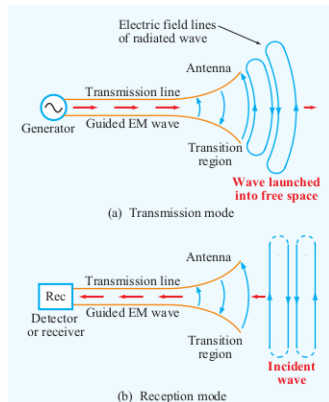
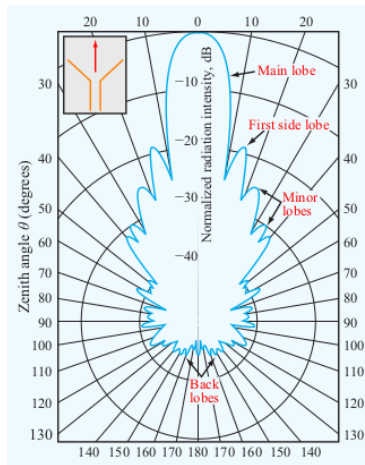


Figure 9-1 Antenna as a transducer between a guided electromagnetic wave and a free-space wave, for both transmission and reception.

Last Lecture (Continued)

Some ways to **characterize** an antenna:

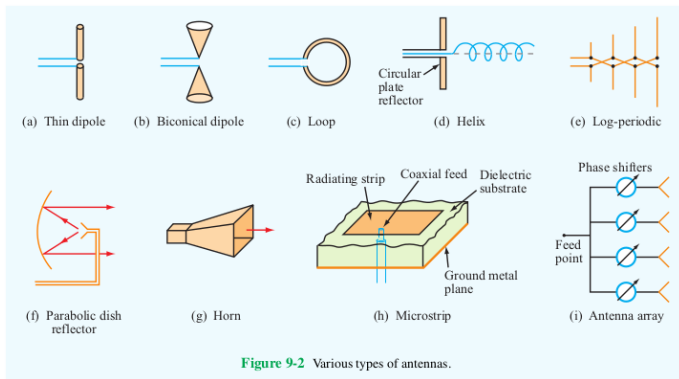
- **antenna radiation pattern** – The **spatial distribution of relative power** that the antenna transmits. Also called the **radiation pattern** or **antenna pattern**.
- **antenna polarization** – The polarization of the emitted wave when the antenna transmits.
- **antenna impedance** associated with the transfer of power from the wave generator to the antenna.



The **relative sensitivity as a receiver** to incoming EM waves from all directions usually **matches its antenna pattern**, which is called **reciprocity**. All antennas considered are reciprocal.

Last Lecture (Continued)

Isotropic Antenna – Hypothetical antenna that radiates (or receives) **uniformly in all directions**, used as a reference.



Antennas are characterized by their radiation source:

- **current sources** – time-varying currents produce the EM waves
- **aperture fields** – EM fields generated across an opening/aperture

Last Lecture (Continued)

Antenna Array – Acts like a single antenna, where the **antenna pattern can be adjusted** to produce a **specific shape** by controlling the waves emitted by each antenna.

Far field – Region **far enough away** from an antenna such that the **wavefronts are planar**. Only investigated the far field.

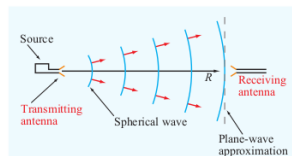


Figure 9-3 Far-field plane-wave approximation.

Hertzian Dipole

Thin, linear conductor with length $l \ll \lambda$ ($l \leq \lambda/50$), where λ is the wavelength of the EM waves it is intended to produce.

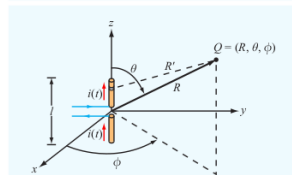


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

Last Lecture (Continued)

The current in a Hertzian dipole is

$$i(t) = I_0 \cos \omega t \text{ (A)} = \text{Re} \{ I_0 e^{j\omega t} \}. \quad (1)$$

So,

$$\tilde{I} = I_0. \quad (2)$$

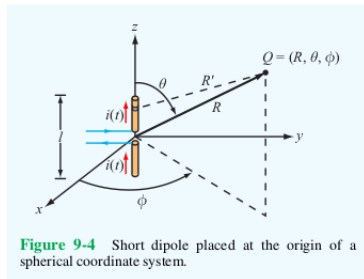
In the **far field**, where $R \gg \lambda$,

$$\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0 \quad (3)$$

$$\tilde{E}_\theta = \frac{jI_0 \ell k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta = \frac{jk\eta_0}{4\pi} \tilde{I}(z) \ell \left(\frac{e^{-jkR}}{R} \right) \sin \theta \text{ (V/m)} \quad (4)$$

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \text{ (A/m)} \quad (5)$$

and the EM wave is a TEM plane wave propagating in the $\hat{\mathbf{R}}$ direction.



Last Lecture (Continued)

The **average power density** that a Hertzian dipole emits is

$$S(R, \theta) = S_0 \sin^2 \theta \quad (\text{W/m}^2), \quad (6)$$

where

$$S_{\max} = S_0 = \frac{\eta_0 k^2 I_0^2 \ell^2}{32\pi^2 R^2} \quad (7)$$

is the maximum power density at a given distance R .

The **normalized radiation intensity**

$$F(\theta, \phi) = \frac{S(R, \theta, \phi)}{S_{\max}} = F(\theta) = \sin^2 \theta \quad (8)$$

specifies the antenna pattern.

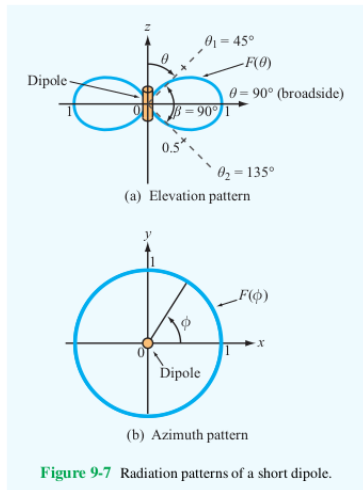


Figure 9-7 Radiation patterns of a short dipole.

Last Lecture (Continued)

Antenna Radiation Characteristics

An **antenna pattern** represents the **far-field directional properties** of an antenna at a given distance. It's the average power that the antenna radiates through a sphere centered on the antenna in the far-field.

- **Total Radiated Power**

$$P_{\text{rad}} = R^2 S_{\text{max}} \iint_{4\pi} F(\theta, \phi) d\Omega \quad (9)$$

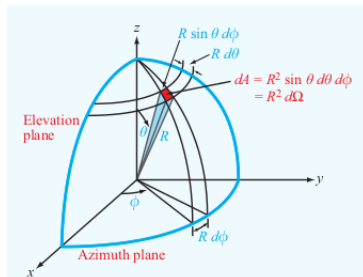


Figure 9-8 Definition of solid angle $d\Omega = \sin \theta d\theta d\phi$.

Last Lecture (Continued)

- **Antenna Pattern** – normalized radiation intensity $F(\theta, \phi)$
 - Can be plotted in 3D.
 - Can be plotted as 2D cross-sections.
 - elevation plane:** ϕ is constant
 - azimuth plane:** θ is constant

Usually plotted on a **decibel scale**
 $F \text{ (dB)} = 10 \log F$.

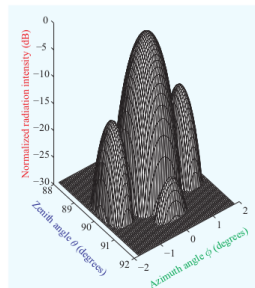
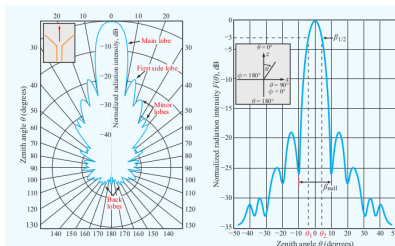
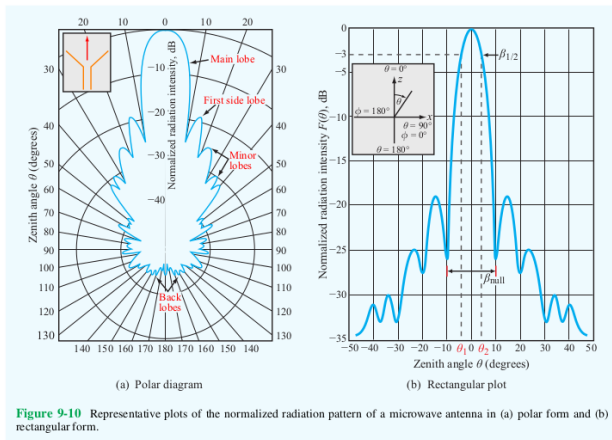


Figure 9-9 Three-dimensional pattern of a narrow-beam antenna.



Last Lecture (Continued)



- **main lobe** – peak containing the largest amount of energy
- **side lobes** – other less prominent peaks in the antenna pattern
- **back lobes** – peaks that point away from the main lobe

Last Lecture (Continued)

• Beam Dimensions

• Pattern Solid Angle

$$\Omega_p = \iint_{4\pi} F(\theta, \phi) d\Omega \quad (\text{sr}) \quad (10)$$

Smaller Ω_p = **more directional**.

Largest Ω_p is 4π , which is the value for an isotropic antenna.

• Beamwidth

half-power or 3 dB beamwidth β

– angular width between points where the power in the main lobe is half its maximum.

null beamwidth β_{null} – angular width between first nulls on opposite sides of the peak

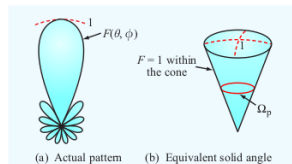
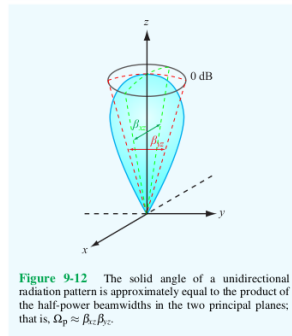


Figure 9-11 The pattern solid angle Ω_p defines an equivalent cone over which all the radiation of the actual antenna is concentrated with uniform intensity equal to the maximum of the actual pattern.



Last Lecture (Continued)

- **Antenna Directivity**

$$D = \frac{F_{\max}}{F_{\text{av}}} = \frac{S_{\max}}{S_{\text{av}}} = \frac{4\pi}{\Omega_p} \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \quad (11)$$

Larger D means more directional. Min value = 1 for isotropic antenna.

- **Antenna Gain**

The **radiation efficiency** of an antenna is

$$\xi = \frac{P_{\text{rad}}}{P_t}, \quad (12)$$

where P_t is the power that the generator applies to the antenna.

Then the antenna **gain**, which accounts for lost power, is

$$G = \frac{4\pi R^2 S_{\max}}{P_t} = \xi D. \quad (13)$$

For a lossless antenna, $\xi = 1$ and $G = D$.

- **Radiation Resistance**

$$R_{\text{rad}} = \frac{2}{I_0^2} P_{\text{rad}} \quad (14)$$

Also, the **radiation efficiency** can be written as

$$\xi = \frac{P_{\text{rad}}}{P_t} = \frac{P_{\text{rad}}}{P_{\text{rad}} + P_{\text{loss}}} = \frac{R_{\text{rad}}}{R_{\text{rad}} + R_{\text{loss}}}, \quad (15)$$

where

$$P_t = P_{\text{rad}} + P_{\text{loss}}, \quad R_{\text{in}} = R_{\text{rad}} + R_{\text{loss}}, \quad \text{and} \quad R_{\text{loss}} = \frac{2}{I_0^2} P_{\text{loss}}. \quad (16)$$

Half-Wave Dipole Antenna

The properties of the **Hertzian dipole**, which has length $\ell \ll \lambda$, can be used to determine the properties of other related antennae. One example is the **half-wave dipole antenna**, which has **length** $\ell = \lambda/2$.

In this case, antenna consists of a **thin wire** with the transmission line connecting at the center. The **current** $i(t)$ in the wire is **0 at the ends** and has the form

$$\begin{aligned} i(t) &= I_0 \cos(\omega t) \cos(kz) \\ &= \text{Re} \left\{ I_0 \cos(kz) e^{j\omega t} \right\} \end{aligned} \quad (17)$$

with **phasor**

$$\tilde{I}(z) = I_0 \cos(kz), \quad (18)$$

where $-\lambda/4 \leq z \leq \lambda/4$
and the **wavenumber** $k = 2\pi/\lambda$.

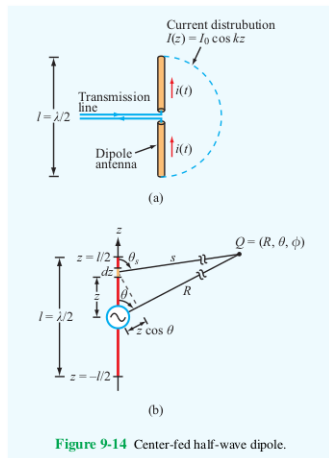


Figure 9-14 Center-fed half-wave dipole.

Half-Wave Dipole Antenna (Continued)

The phasor $d\tilde{E}_\theta$ for the θ component of the electric field for a dipole segment of length dz at position z in the dipole and at a distance s from the segment can be determined using the expression for a Hertzian dipole.

For a Hertzian dipole, with current phasor $\tilde{I}(z) = I_0$,

$$\tilde{E}_\theta = \frac{jI_0 \ell k \eta_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta = \frac{jk\eta_0}{4\pi} \tilde{I}(z) \ell \left(\frac{e^{-jkR}}{R} \right) \sin \theta. \quad (19)$$

The other two components $\tilde{E}_R = \tilde{E}_\phi = 0$. The only nonzero component of the magnetic field phasor is $\tilde{H}_\phi = \tilde{E}_\theta / \eta_0$. So, **for the dipole segment**,

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left(\frac{e^{-jks}}{s} \right) \sin \theta_s \quad (20)$$

and

$$d\tilde{H}_\phi(z) = d\tilde{E}_\theta(z) / \eta_0 \quad (21)$$

with all other components = 0.

Half-Wave Dipole Antenna (Continued)

The **electric field in the far field** is

$$\tilde{E}_\theta = \int_{z=-\lambda/4}^{\lambda/4} d\tilde{E}_\theta(z). \quad (22)$$

Assuming

- $1/s \approx 1/R$
- $\theta_s \approx \theta$
- $e^{jks} \approx e^{jk(R-z \cos \theta)}$,

$$d\tilde{E}_\theta(z) = \frac{jk\eta_0}{4\pi} \tilde{I}(z) dz \left(\frac{e^{-jkR}}{R} \right) \sin \theta e^{jkz \cos \theta} \quad (23)$$

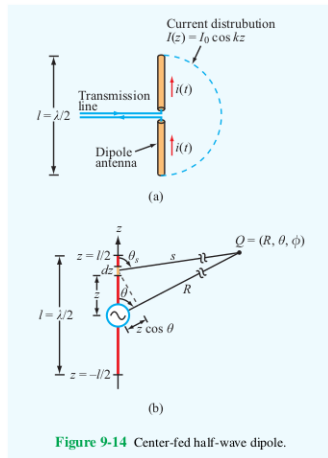


Figure 9-14 Center-fed half-wave dipole.

Half-Wave Dipole Antenna (Continued)

and the **phasors of the electric and magnetic fields** are

$$\begin{aligned}\tilde{E}_\theta &= \frac{jk\eta_0 I_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \int_{z=-\lambda/4}^{\lambda/4} \cos(kz) e^{jkz \cos \theta} dz \\ &= j60 I_0 \left(\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right) \left(\frac{e^{-jkR}}{R} \right)\end{aligned}\quad (24)$$

and

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \quad (25)$$

with $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$.

The **average power density** is

$$\begin{aligned}S(R, \theta) &= \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{1}{240\pi} 3600 I_0^2 \left(\frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta} \right) \left(\frac{1}{R^2} \right) \\ &= \frac{15 I_0^2}{\pi R^2} \left(\frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta} \right) = S_0 \left(\frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta} \right),\end{aligned}\quad (26)$$

Half-Wave Dipole Antenna (Continued)

where

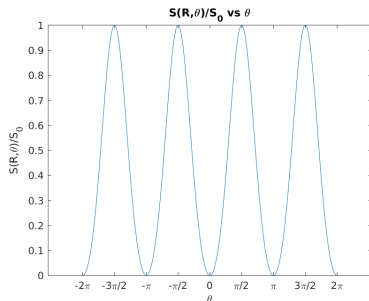
$$S_0 = \frac{15I_0^2}{\pi R^2}. \quad (27)$$

The factor

$$\frac{S(R, \theta)}{S_0} = \left(\frac{\cos^2 \left(\frac{\pi}{2} \cos \theta \right)}{\sin^2 \theta} \right) \quad (28)$$

is periodic with period π and reaches its maximum value of 1 at $\theta = \pi/2$. It also has removable singularities when $\sin \theta = 0$. (These can be shown using the usual methods from calculus.) Therefore,

$$S_{\max} = S_0 \quad (29)$$



Half-Wave Dipole Antenna (Continued)

and the **normalized radiation intensity** is

$$F(\theta, \phi) = F(\theta) = S(R, \theta)/S_0 = \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2. \quad (30)$$

Directivity

The **total radiated power** of the half-wave dipole antenna is

$$\begin{aligned} P_{\text{rad}} &= R^2 \int \int_{4\pi} S(R, \theta) d\Omega = \frac{15I_0^2}{\pi} \int_0^{2\pi} \int_0^\pi \left(\frac{\cos\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} \right)^2 \sin \theta d\theta d\phi \\ &= 30I_0^2 \int_0^\pi \frac{\cos^2\left(\frac{\pi}{2} \cos \theta\right)}{\sin \theta} d\theta \quad (\text{W}). \end{aligned} \quad (31)$$

The integral can be evaluated numerically and has the value 1.22 so that

$$P_{\text{rad}} = 30I_0^2(1.22) = 36.6I_0^2 \quad (\text{W}). \quad (32)$$

Half-Wave Dipole Antenna (Continued)

So, the **directivity** of the half-wave dipole antenna is

$$D = \frac{4\pi R^2 S_{\max}}{P_{\text{rad}}} = \frac{60 I_0^2}{36.6 I_0^2} = 1.64. \quad (33)$$

In terms of decibels,

$$D \text{ (dB)} = 10 \log 1.64 = 2.15 \text{ dB}. \quad (34)$$

Radiation Resistance

The **radiation resistance** of the half-wave dipole antenna is

$$R_{\text{rad}} = \frac{2}{I_0^2} P_{\text{rad}} = 73.2 \text{ } (\Omega). \quad (35)$$

It's easier to match a transmission line to this resistance than it is to match a transmission line to a Hertzian dipole antenna with $R_{\text{rad}} \sim 0.1 \text{ } \Omega$.

Half-Wave Dipole Antenna (Continued)

Also, for a half-wave dipole antenna, $R_{\text{loss}} \ll R_{\text{rad}}$ so that

$$R_{\text{in}} \approx R_{\text{rad}} \quad (36)$$

and the **input impedance**

$$Z_{\text{in}} \approx R_{\text{rad}} + jX_{\text{in}}. \quad (37)$$

Here, the **reactive component** X_{in} has a strong dependence on ℓ/λ . Finding an explicit expression is beyond our scope but

$$X_{\text{in}} = \begin{cases} 42 \, \Omega, & \ell/\lambda = 0.5 \\ 0 \, \Omega, & \ell/\lambda = 0.48. \end{cases} \quad (38)$$

So, when the length ℓ is decreased by a small amount, $Z_{\text{in}} = 73.2 \, \Omega \in \mathbb{R}$ and the antenna can be matched with a $75 \, \Omega$ transmission line without a matching network.

Half-Wave Dipole Antenna (Continued)

Quarter-Wave Monopole Antenna

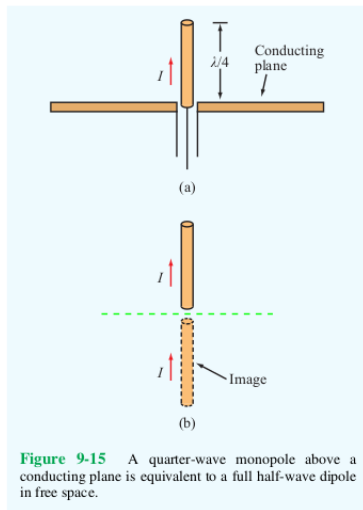
Using **image theory**, the properties of a **quarter-wave monopole antenna** above a conducting ground plane can be inferred from the properties of a half-wave dipole antenna. As before, the electric and magnetic field phasors are

$$\tilde{E}_\theta = j60I_0 \left(\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} \right) \left(\frac{e^{-jkR}}{R} \right) \quad (39)$$

$$\tilde{H}_\phi = \tilde{E}_\theta / \eta_0 \quad (40)$$

with $\tilde{E}_R = \tilde{E}_\phi = \tilde{H}_R = \tilde{H}_\theta = 0$.

For the monopole antenna, this field is



Half-Wave Dipole Antenna (Continued)

only present in the upper half-space with

$$0 \leq \theta \leq \pi/2, \quad (41)$$

the **normalized radiation intensity** is

$$F(\theta) = \left(\frac{\cos(\frac{\pi}{2} \cos \theta)}{\sin \theta} \right)^2, \quad (42)$$

the antenna only emits **half the power** of the dipole antenna

$$P_{\text{rad}} = \frac{1}{2}(36.6 I_0^2) = 18.3 I_0^2 \text{ (W)}, \quad (43)$$

and the **radiation resistance** is

$$R_{\text{rad}} = \frac{1}{2}(73.2) = 36.6 \text{ } (\Omega). \quad (44)$$

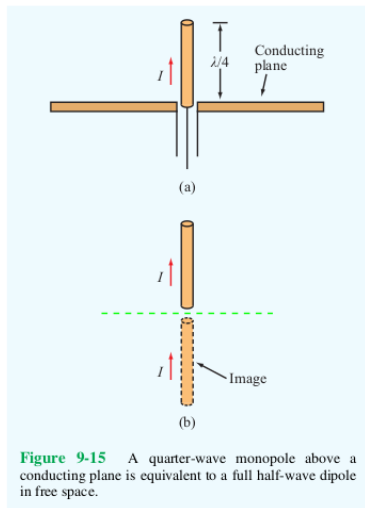


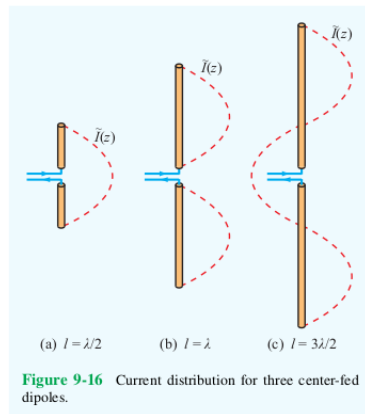
Figure 9-15 A quarter-wave monopole above a conducting plane is equivalent to a full half-wave dipole in free space.

Arbitrary-Length Dipole Antennas

For a **center-fed dipole**, the currents through the two halves are symmetrical and 0 at the ends $z = -\ell/2, 0, \ell/2$:

$$\tilde{I}(z) = I_0 \begin{cases} \sin(k(\frac{\ell}{2} - z)), & 0 \leq z \leq \frac{\ell}{2} \\ \sin(k(\frac{\ell}{2} + z)), & -\frac{\ell}{2} \leq z < 0 \end{cases} \quad (45)$$

where I_0 is the constant amplitude. As before, the expressions for the **electric and magnetic field phasors of the Hertzian dipole** can be used to determine the electric and magnetic field phasors in this case.



Arbitrary-Length Dipole Antennas (Continued)

In this case, assuming

- $1/s \approx 1/R$
- $\theta_s \approx \theta$
- $e^{jks} \approx e^{jk(R-z \cos \theta)}$,

$$d\tilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta e^{jkz \cos \theta} dz \begin{cases} \sin(k(\frac{\ell}{2} - z)), & 0 \leq z \leq \frac{\ell}{2} \\ \sin(k(\frac{\ell}{2} + z)), & -\frac{\ell}{2} \leq z < 0 \end{cases} \quad (46)$$

and

$$\begin{aligned} \tilde{E}_\theta &= \int_{z=-\ell/2}^{\ell/2} d\tilde{E}_\theta = \int_{z=-\ell/2}^0 d\tilde{E}_\theta + \int_{z=0}^{\ell/2} d\tilde{E}_\theta = \frac{jk\eta_0 I_0}{4\pi} \left(\frac{e^{-jkR}}{R} \right) \sin \theta \\ &\quad \cdot \left(\int_{-\ell/2}^0 e^{jkz \cos \theta} \sin \left(k \left(\frac{\ell}{2} + z \right) \right) dz \right. \\ &\quad \left. + \int_0^{\ell/2} e^{jkz \cos \theta} \sin \left(k \left(\frac{\ell}{2} - z \right) \right) dz \right). \quad (47) \end{aligned}$$

Arbitrary-Length Dipole Antennas (Continued)

Here,

$$\begin{aligned} & \int e^{jkz \cos \theta} \sin \left(k \left(\frac{\ell}{2} \pm z \right) \right) dz \\ &= \int \cos(kz \cos \theta) \sin \left(k \left(\frac{\ell}{2} \pm z \right) \right) dz \\ & \quad + j \int \sin(kz \cos \theta) \sin \left(k \left(\frac{\ell}{2} \pm z \right) \right) dz. \end{aligned} \quad (48)$$

so that the **nonzero component of the electric field phasor** is

$$\tilde{E}_\theta = j60I_0 \left(\frac{e^{-jkR}}{R} \right) \left(\frac{\cos \left(\frac{k\ell}{2} \cos \theta \right) - \cos \left(\frac{k\ell}{2} \right)}{\sin \theta} \right). \quad (49)$$

Using $k = 2\pi/\lambda$, the **average power density** of the antenna is

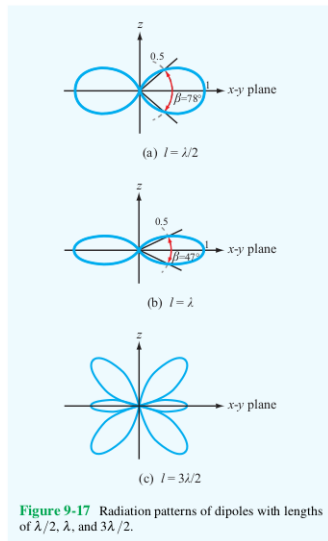
$$S(R, \theta) = \frac{|\tilde{E}_\theta|^2}{2\eta_0} = \frac{15I_0^2}{\pi R^2} \left(\frac{\cos \left(\frac{\pi\ell}{\lambda} \cos \theta \right) - \cos \left(\frac{\pi\ell}{\lambda} \right)}{\sin \theta} \right)^2. \quad (50)$$

Arbitrary-Length Dipole Antennas (Continued)

The **antenna patterns** when

$$\ell = \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2} \quad (51)$$

are shown in the adjacent figure. There is also a module accessible through the companion website that allows you to interactively explore the characteristics of this type of antenna.



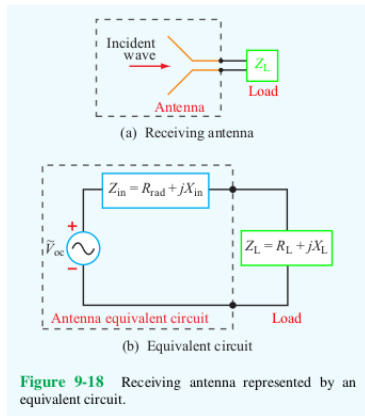
Effective Area of a Receiving Antenna

The quantities used to characterize an antenna as an **EM wave receiver** differ from those used when it's an **EM wave emitter**.

If an incident EM wave has power density S_i (W/m^2) and the antenna collects an **intercepted power** P_{int} (W), the **effective area**, **effective aperture**, or **receiving cross section** of the antenna is defined as

$$A_e = \frac{P_{\text{int}}}{S_i} \text{ (m}^2\text{)}. \quad (52)$$

The **circuit** in the figure represents the transfer of power through the antenna to the load.



Effective Area of a Receiving Antenna (Continued)

The incident wave induces the **open-circuit voltage** $\tilde{V}_{oc}$ at the antenna terminals, Z_{in} is the **input impedance** of the antenna, and Z_L is the **impedance of the load**. In this case, assuming $R_{loss} \ll R_{rad}$,

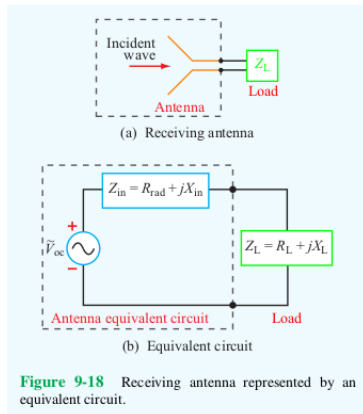
$$Z_{in} = R_{rad} + jX_{in}, \quad (53)$$

where R_{rad} is the **radiation resistance**, and

$$Z_L = R_L + jX_L. \quad (54)$$

Ensuring that

$$Z_L = Z_{in}^*, \quad (55)$$



Effective Area of a Receiving Antenna (Continued)

i.e.

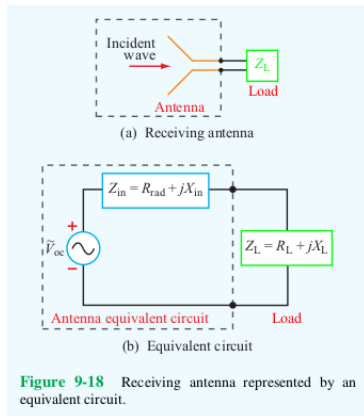
$$R_L = R_{\text{rad}} \quad \text{and} \quad X_L = -X_{\text{in}}, \quad (56)$$

will maximize power transfer to the load.

When $Z_L = Z_{\text{in}}^*$, the overall impedance is $2R_{\text{rad}}$ and it's connected to the sinusoidal voltage source $\tilde{V}_{\text{oc}}$. This means that the **power supplied to the load** is

$$\begin{aligned} P_L &= \frac{1}{2} |\tilde{I}_L|^2 R_{\text{rad}} = \frac{1}{2} \left(\frac{|\tilde{V}_{\text{oc}}|}{2R_{\text{rad}}} \right)^2 R_{\text{rad}} \\ &= \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}, \end{aligned} \quad (57)$$

where $\tilde{I}_L$ is the current in the circuit.



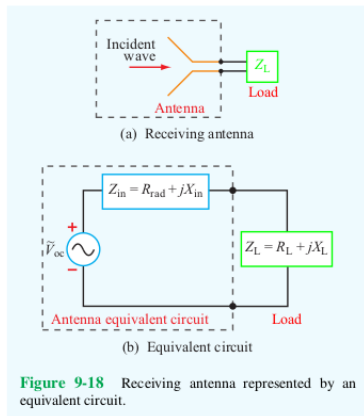
Effective Area of a Receiving Antenna (Continued)

All of the intercepted power is transferred to the load, which means

$$P_{\text{int}} = P_L = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}}. \quad (58)$$

If the electric field $\tilde{E}_i$ is the component of the incident wave **parallel to the polarization direction** of the antenna, then the power density of the wave is

$$S_i = \frac{|\tilde{E}_i|^2}{2\eta_0} = \frac{|\tilde{E}_i|^2}{240\pi}. \quad (59)$$



Effective Area of a Receiving Antenna (Continued)

So, the **effective area of the antenna** is

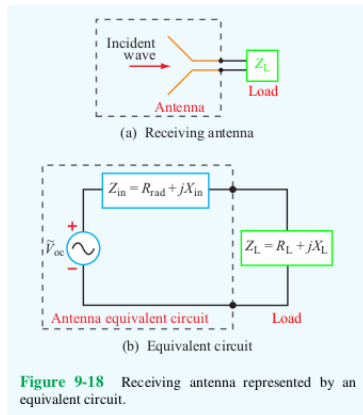
$$\begin{aligned} A_e &= \frac{P_{\text{int}}}{S_i} = \frac{|\tilde{V}_{\text{oc}}|^2}{8R_{\text{rad}}} \left(\frac{240\pi}{|\tilde{E}_i|^2} \right) \\ &= \frac{30\pi |\tilde{V}_{\text{oc}}|^2}{R_{\text{rad}} |\tilde{E}_i|^2}. \end{aligned} \quad (60)$$

For any antenna, this formula implies

$$A_e = \frac{\lambda^2 D}{4\pi} \text{ (m}^2\text{)}. \quad (61)$$

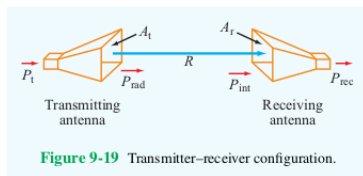
In particular, for a **Hertzian dipole**,

$$A_e = \frac{3\lambda^2}{8\pi} \text{ (m}^2\text{)}. \quad (62)$$



Friis Transmission Formula

Consider the two antennas in the figure, where the **antenna on the left**, with effective area A_t and radiation efficiency ξ_t , **emits EM waves** and the **antenna on the right**, with effective area A_r and radiation efficiency ξ_r , **receives them**. Suppose that the antennas are oriented so that the **peaks of their antenna patterns are aligned** and that the transmission lines are impedance matched to the antennas. Let P_t be the **power applied to the transmitting antenna** and P_{rec} be the **power received by the receiver**.



Friis Transmission Formula (Continued)

A **lossless isotropic radiator** emits all of the power applied to the antenna evenly in all directions so that the average power density of the radiator is

$$S_{\text{iso}} = \frac{P_t}{4\pi R^2}. \quad (63)$$

Then the **incident power density at the receiving antenna** is

$$S_r = G_t S_{\text{iso}} = \xi_t D_t S_{\text{iso}} = \frac{\xi_t D_t P_t}{4\pi R^2}. \quad (64)$$

By solving the formula for the effective area A_t of the transmitting antenna for the directivity D_t ,

$$D_t = \frac{4\pi A_t}{\lambda^2} \quad (65)$$

and

$$S_r = \frac{\xi_t A_t P_t}{\lambda^2 R^2}. \quad (66)$$

Friis Transmission Formula (Continued)

So, the **power that arrives at the receiver** is

$$P_{\text{int}} = S_r A_r = \frac{\xi_t A_t A_r P_t}{\lambda^2 R^2}. \quad (67)$$

Finally, the **power received by the receiver** is

$$P_{\text{rec}} = \xi_r P_{\text{int}}, \quad (68)$$

$$A_t = \frac{\lambda^2 D_t}{4\pi} = \frac{\lambda^2 G_t}{4\pi \xi_t}, \quad \text{and} \quad A_r = \frac{\lambda^2 D_r}{4\pi} = \frac{\lambda^2 G_r}{4\pi \xi_r} \quad (69)$$

so that the **power transfer ratio** is

$$\frac{P_{\text{rec}}}{P_t} = \frac{\xi_t \xi_r A_t A_r}{\lambda^2 R^2} = G_t G_r \left(\frac{\lambda}{4\pi R} \right)^2. \quad (70)$$

This is called the **Friis transmission formula**.

Friis Transmission Formula (Continued)

More generally, if the peaks of the antenna patterns are not aligned,

$$\frac{P_{\text{rec}}}{P_{\text{t}}} = G_{\text{t}} G_{\text{r}} \left(\frac{\lambda}{4\pi R} \right)^2 F_{\text{t}}(\theta_{\text{t}}, \phi_{\text{t}}) F_{\text{r}}(\theta_{\text{r}}, \phi_{\text{r}}). \quad (71)$$

Noise Power and the Signal-To-Noise Ratio

One way to characterize the noise in a communication system is based on the **system noise temperature** T_{sys} (kelvins (K)), which is a measure of

- the EM noise the antenna receives
- and the noise the electronics generate.

Using this value, the **noise power** is

$$P_n = KT_{\text{sys}}B \text{ (W)}, \quad (72)$$

where

- $K = 1.38 \times 10^{-23} \text{ J/K}$ is **Boltzmann's constant**
- B is the **receiver bandwidth** in Hz.

The **signal-to-noise ratio (SNR)** is

$$S_n = P_{\text{rec}}/P_n. \quad (73)$$

Note: There are other types of noise and other definitions for SNR.

Homework

Read Chapter 9.

Homework 3 is posted and due Tuesday at the beginning of class.

Exam 2 will be on Tuesday, June 9, from 1:30 – 3:20 pm.
(Please verify this and let me know if I'm wrong.)